

If $A^T A = (C_{ij}) \in \mathbb{R}^{n \times n}$, then $C_{ij} = \underbrace{\text{"the } i\text{-th row of } A^T"}_{A^T} \cdot \underbrace{\text{"the } j\text{-th column of } A"}_A$

the i th row
of $A^T \rightarrow \begin{pmatrix} a_{1i} & a_{2i} & \dots & a_{ni} \end{pmatrix} \cdot \begin{pmatrix} a_{1j} \\ a_{2j} \\ a_{3j} \\ \vdots \\ a_{mj} \end{pmatrix}$

↑ the j -th column
of A

$$= a_{1i} a_{1j} + a_{2i} a_{2j} + \dots + a_{mi} a_{mj}$$

$$= \sum_{k=1}^m a_{ki} a_{kj}$$

In particular: $C_{ii} = a_{i1} a_{i1} + a_{i2} a_{i2} + \dots + a_{in} a_{in}$

$$= \sum_{k=1}^m a_{ki} a_{kj}$$

$$= \sum_{k=1}^3 a_{ki}^2$$

(This would be helpful for Homework 1, problem 3)

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5). Conjugate transpose of a matrix in $\mathbb{C}^{m \times n}$

$$A = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix} \in \mathbb{C}^{m \times n}, \text{ then } A^* = \begin{pmatrix} \bar{a}_{11} & \bar{a}_{21} & \dots & \bar{a}_{m1} \\ \bar{a}_{12} & \bar{a}_{22} & \dots & \bar{a}_{m2} \\ \vdots & \vdots & \ddots & \vdots \\ \bar{a}_{1n} & \bar{a}_{2n} & \dots & \bar{a}_{mn} \end{pmatrix} \in \mathbb{C}^{n \times m}$$

(Recall let $z = a + bi \in \mathbb{C}$ with $a, b \in \mathbb{R}$, then $\bar{z} = a - bi$ is the conjugate of z ..

Note. $z \cdot \bar{z} = \bar{z} \cdot z = a^2 + b^2$.)

More examples:

- 4). Symmetric matrix: $A \in \mathbb{R}^{n \times n}$ is called symmetric if $A = A^T$
- 5) Hermitian matrix: $A \in \mathbb{C}^{n \times n}$ is called Hermitian if $A = A^*$

Determinant of a square matrix:

$$A = (a_{ij}) \in \mathbb{R}^{n \times n} \quad (\text{or } \mathbb{C}^{n \times n})$$

$$\det A = \sum_{j=1}^n (-1)^{i+j} a_{ij} \det(M_{ij})$$

Where M_{ij} is the (i,j) -th minor of A , which is an $(n-1) \times (n-1)$

submatrix of A obtained by deleting the i -th row and j -th column of A .

Invertible matrix:

Let $A \in \mathbb{R}^{n \times n}$ (or $\mathbb{C}^{n \times n}$) is invertible if there exists $B \in \mathbb{R}^{n \times n}$ (or $\mathbb{C}^{n \times n}$)

such that $AB = BA = I_n$.

Notation: $B = A^{-1}$

Properties of determinants and inverses.

Let $A \in \mathbb{R}^{n \times n}$ (or $\mathbb{C}^{n \times n}$).

1). If the rows (or columns) of A are linearly dependent, then $\det(A) = 0$

2). A is invertible $\Leftrightarrow \det A \neq 0$

3). $\det A = \det A^T$ if $A \in \mathbb{R}^{n \times n}$; $\det(A) = \overline{\det A^*}$, if $A \in \mathbb{C}^{n \times n}$.

4). $\det(AB) = \det A \cdot \det B = \det(BA)$.

5). If A is invertible, then $\det(A^{-1}) = \frac{1}{\det A}$.

Null space (or kernel) and range space.

$$\text{Let } A = \begin{pmatrix} \frac{1}{a_1} & \frac{1}{a_2} & \dots & \frac{1}{a_n} \end{pmatrix} \in \mathbb{R}^{m \times n} \text{ (or } \mathbb{C}^{m \times n})$$

$$\text{Range } A = \text{Span} \{ \vec{a}_1, \dots, \vec{a}_n \} = \{ c_1 \vec{a}_1 + \dots + c_n \vec{a}_n : \forall c_1, \dots, c_n \in \mathbb{R} \text{ (or } \mathbb{C}) \}$$

$$\text{Rank } A = \dim(\text{Range } A)$$

$$\text{Null}(A) = \text{Ker}(A) = \{ \vec{x} \in \mathbb{R}^n \text{ (or } \mathbb{C}^n) : A\vec{x} = \vec{0} \}$$

$$\text{nullity}(A) = \dim(\text{Null}(A)).$$

Example. Given that $A = \begin{pmatrix} 1 & -3 & 4 \\ 3 & -7 & 7 \\ -4 & 6 & -1 \end{pmatrix}$ has reduced echelon form $\begin{pmatrix} \textcircled{1} & 0 & -7/2 \\ 0 & \textcircled{1} & -5/2 \\ 0 & 0 & 0 \end{pmatrix}$ ↓ pivots

Find a basis for $\text{Null}(A)$ and a basis for $\text{Range}(A)$.

Answer: A basis for range of A : $\left\{ \begin{pmatrix} 1 \\ 3 \\ -4 \end{pmatrix}, \begin{pmatrix} -3 \\ -7 \\ 6 \end{pmatrix} \right\}$ these are pivot columns of A .

$$\text{Null}(A) = \text{Solution set of } A\vec{x} = \vec{0}$$

$$\text{Therefore: } A\vec{x} = \vec{0} \Leftrightarrow \begin{cases} x_1 - 7/2 x_3 = 0 \\ x_2 - 5/2 x_3 = 0 \end{cases} \Leftrightarrow \begin{cases} x_1 = 7/2 x_3 \\ x_2 = 5/2 x_3 \\ x_3 = x_3 \end{cases} \Leftrightarrow \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 7/2 \\ 5/2 \\ 1 \end{pmatrix} x_3$$

$$\Rightarrow \text{A basis for Null}(A) \text{ is given by } \left\{ \begin{pmatrix} 7/2 \\ 5/2 \\ 1 \end{pmatrix} \right\}.$$

$$\text{Rank-nullity Theorem: } \text{Rank}(A) + \text{nullity}(A) = n = \# \text{ of columns of } A.$$

Propositions: Let $A \in \mathbb{R}^{n \times n}$ (or $\mathbb{C}^{n \times n}$). Then the following statements are equivalent

- 1) A is invertible
- 2) $\det A \neq 0$
- 3) $\text{Null}(A) = \{ \vec{0} \}$
- 4) $\text{Rank}(A) = n$ (or $\mathbb{C}^n$).

Other reviews: • inner products

• eigenvalues, eigenvectors, eigenspace, diagonalization.

Vector spaces, subspaces, bases and dimensions.

Vector Spaces

Definition: A vector space V over the field of $\mathbb{R}$ (or $\mathbb{C}$) is a collection of objects (called vectors), together with an operation (referred as addition "+") and scalar multiplication over $\mathbb{R}$ (or $\mathbb{C}$) such that the axiom holds for all vectors $\vec{x}, \vec{y}$, and $\vec{z} \in V$ and for any scalar $a, b \in \mathbb{R}$ (or $\mathbb{C}$).

0). $\vec{x} + \vec{y} \in V, a\vec{x} \in V$ i.e. V is closed under "+" and the scalar multiplication

I). $\vec{x} + \vec{y} = \vec{y} + \vec{x}$ i.e. "+" is commutative

II). $(\vec{x} + \vec{y}) + \vec{z} = \vec{x} + (\vec{y} + \vec{z})$ i.e. "+" is associative

III). There exists a vector $\vec{0} \in V$ such that $\vec{x} + \vec{0} = \vec{0} + \vec{x} = \vec{x}$

(Homework: Show that the zero vector is unique).

IV). For each $\vec{x} \in V$, there exists $\vec{v} \in V$ such that $\vec{x} + \vec{v} = \vec{0}$.

v). $1\vec{x} = \vec{x}$

VI). $(a \cdot b)\vec{x} = a(b\vec{x})$, i.e. scalar multiplication is associative

VII). $(a+b)\vec{x} = a\vec{x} + b\vec{x}$

VIII). $a(\vec{x} + \vec{y}) = a\vec{x} + a\vec{y}$

} "+" and scalar multiplication is distributive.

Examples of Vector Spaces:

- 1). $\mathbb{R}^n$ or $\mathbb{C}^n$ with "+" and scalar multiplication
- 2). $\mathbb{R}^{m \times n}$ or $\mathbb{C}^{m \times n}$ with matrix addition and scalar multiplication
- 3). $P(\mathbb{R})$ or $P(\mathbb{C})$ = the set of all polynomials with real/complex coefficients

"+" : polynomial addition by "combining like terms"

scalar multiplication: $\forall f(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$

$$cf(x) = (ca_n)x^n + (ca_{n-1})x^{n-1} + \dots + (ca_1)x + ca_0.$$

- 4). $P^n(\mathbb{R})$ or $P^n(\mathbb{C})$ = the set of polynomials with degrees less or equal to n .

Question: Let S be the set of polynomials with degrees equal to

n . Is S a vector space? Explain.

- 5). $C^0(0,1)$ = the set of all continuous function on $(0,1)$.

"+" : function addition: $\forall f$ and $g \in C^0(0,1)$

$$(f+g)(x) = f(x) + g(x).$$

scalar multiplication: $\forall c \in \mathbb{R}$ and $f \in C^0(0,1)$

$$(cf)(x) = c(f(x)).$$

Example: Let $V = \{(a_1, a_2) : a_1, a_2 \in \mathbb{R}\}$. $\forall (a_1, a_2), (b_1, b_2) \in V$ and $c \in \mathbb{R}$.

Define 1) : "+" : $(a_1, a_2) + (b_1, b_2) = (a_1 + b_1, a_2 + b_2)$ and

$$2) : \text{scalar multiplication: } c(a_1, a_2) = \begin{cases} (0, 0), & \text{if } c = 0 \\ (ca_1, \frac{a_2}{c}), & \text{if } c \neq 0. \end{cases}$$

Is V a vector space with the above "+" and scalar multiplication? Explain.

Answer: V is not a vector space because VII fails.

Counter-example to VII :

$$\text{Denote } \vec{x} = (x_1, x_2). \quad (a+b)\vec{x} = ((a+b)x_1, \frac{x_2}{a+b})$$

$$\begin{aligned} a\vec{x} + b\vec{x} &= (ax_1, \frac{x_2}{a}) + (bx_1, \frac{x_2}{b}) \\ &= ((a+b)x_1, \frac{x_2}{a} + \frac{x_2}{b}) \end{aligned}$$

$$\text{In general, } \frac{x_2}{a+b} = \frac{x_2}{a} + \frac{x_2}{b} \quad \text{for example take } a=b=x_2=1$$

$$\text{Then: } \frac{x_2}{a+b} = \frac{1}{2} \neq \frac{x_2}{a} + \frac{x_2}{b} = 2$$

Subspaces

Definition: Let V be a vector space over $\mathbb{R}$ (or $\mathbb{C}$). A subset W of V is a subspace if W is a non-empty set of V and it itself is a vector space.

Proposition (or equivalent definition): Let V be a vector space over $\mathbb{R}$ (or $\mathbb{C}$)

then $W \subseteq V$ is a subspace if and only if W satisfies the following conditions

1). $W \neq \emptyset$, and

2) $\forall \vec{x}, \vec{y} \in W: \vec{x} + \vec{y} \in W$, and

3) $\forall c \in \mathbb{R}$ (or $\mathbb{C}$) and $\vec{x} \in W: c\vec{x} \in W$.

} W is closed under "+" and scalar multiplication

Remark: 2) and 3) together is equivalent to the following condition:

$$2') \quad \forall \vec{x}, \vec{y} \in W \text{ and } \forall c \in \mathbb{R} \text{ (or } \mathbb{C}\text{): } c\vec{x} + \vec{y} \in W$$

Examples: 1). $P^n(\mathbb{R}) \subseteq P(\mathbb{R})$ is a subspace

2). $S =$ the set of all diagonal matrices in $\mathbb{R}^{n \times n}$ (or $\mathbb{C}^{n \times n}$), $V = \mathbb{R}^{n \times n}$ (or $\mathbb{C}^{n \times n}$)

$\Rightarrow S \subseteq V$ is a subspace.

3). $V = \mathbb{C}$ over $\mathbb{R}$. (meaning the scalars $c \in \mathbb{R}$), $S = \mathbb{R}$

$\Rightarrow S \subseteq V$ is a subspace

4) $V = \mathbb{C}$ over $\mathbb{C}$, $S = \mathbb{R}$

$\Rightarrow S$ is not a subspace.

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5). $V = \mathbb{R}^{n \times n}$, $S =$ the set of all symmetric matrices

$$\text{i.e.: } S = \{A \in V : A^T = A\}$$

(Homework: show that S is a subspace of V .)

6). Definition: Let $S = \{\vec{v}_1, \dots, \vec{v}_k\} \subseteq V$ be a subset. The span of S is

$$\text{Span } S = \{c_1 \vec{v}_1 + c_2 \vec{v}_2 + \dots + c_k \vec{v}_k : c_1, c_2, \dots, c_k \in \mathbb{R} \text{ (or } \mathbb{C})\}.$$

Then $\text{Span } S$ is a subspace of V .

Linear independence / dependent relations

Definition: A finite set of vectors $\{\vec{x}_1, \dots, \vec{x}_k\} \subseteq V$ is called linearly dependent

if there exists $c_1, c_2, \dots, c_k$, which are not all zeros, such that

$$c_1 \vec{x}_1 + c_2 \vec{x}_2 + \dots + c_k \vec{x}_k = \vec{0}.$$

Definition: A finite set of vectors $\{\vec{x}_1, \dots, \vec{x}_k\} \subseteq V$ is called linearly independent

if $c_1 \vec{x}_1 + \dots + c_k \vec{x}_k = \vec{0}$ implies $c_1 = c_2 = \dots = c_k = 0$.

Examples: 1). $\{\vec{x}\} \subseteq V$ is linearly independent if and only if $\vec{x} \neq \vec{0}$.

2) $\{\vec{x}_1, \vec{x}_2\} \subseteq V$ is linearly dependent if and only if $\vec{x}_1$ and $\vec{x}_2$ is scalar multiples of each other